HSLU Lucerne University of Applied Sciences and Arts

**Algorithms and Data Structures for creative technologies (ADCREA)**

# ADCREA Semester Project Requirements

## Overview

The purpose of this semester project is to apply and demonstrate their understanding and application of Data Structures and Algorithms using C# in Unity 6. Students are encouraged to use VR on Quest. Focus is on showcasing their ability to design, implement, and integrate Data Structures and Algorithms.

The project may be done individually or in teams of two students, and it will count together with its presentation for 30% to 70% of the semester grade, depending on the individual selection.

### Key Information

| | | | |
|---|---|---|---|
| Responsible: | Prof. Laszlo Arato | S12.019 | laszlo.arato@hslu.ch |
| Technical Support: | Christian Hirt | | christian.hirt@hslu.ch |
| | Silvan Monteiro | S1A.7 | silvan.monteiro@hslu.ch |
| | Tobias Kreienbühl | S12.018 | tobias.kreienbuel@hslu.ch |
| Project Type: | Semester Project | Section: | BSc. Digital Ideation and BSc. Immersive Technologies |
| Start: | Tuesday, March 31st , 2026 | | During the morning lecture |
| Submission of work: | Monday, June 15th, 2026 | | End of the day, 24:00 |
| Presentation: | Tuesday, June 30th, 2026 and Wednesday, July 1st, 2026 | | Exact times will be scheduled |

Laszlo Arato, revision 1.1 (Spring 2026)

## 1 Project Goal

The purpose of this semester project is to apply and demonstrate their understanding and application of Data Structures and Algorithms using C# in Unity 6. Students are encouraged to use VR on Quest. Focus is on showcasing their ability to design, implement, and integrate Data Structures and Algorithms.

The project may be done individually or in teams of two students, and it will count together with its presentation for 30% to 70% of the semester grade, depending on the individual selection.

The application must **NOT** be a full game, but rather a functional demonstrator!
(This is the beginning of your learning curve as a creative designer – not the end of it!).

# 2 Project Requirements

## 2.1 Data Structure Requirements

The project must include at least three different instances of the following data structures:

- ArrayList ❑
- List ❑
- SortedList ❑
- LinkedList ❑
- Queue ❑
- Stack ❑
- Hash ❑
- Dictionary or ❑
- SortedSet ❑

## 2.2 Algorithm Requirements

The project must include at least three different tree or graph algorithms from the following list:

- Depth First Traversal ❑
- Breadth First Traversal ❑
- Binary Search Tree ❑
- Warshall algorithm to find all vertices in a graph ❑
- Detect cycles in an undirected graph ❑
- Detect cycles in a directed graph ❑
- Floyd-Warshall to find all shortest paths in a densely populated graph ❑
- Dijkstra algorithm to find the shortest path starting at one vertex ❑
- A* algorithm to find the shortest path in a geographical graph ❑
- Any other tree or graph algorithm not listed here ❑ ❑ ❑

## 2.3 Data Story

Select a suitable application with meaningful data in a credible environment solving realistic challenge(s).

Be creative – there are many data driven operations that require some form of optimization or decision consideration or resource allocation or trade tracking …

## 2.4 Visualization Requirements

There must be some form of visualization showing the data structure representation or manipulation.

Manual interaction with the data or the algorithms is highly valued, but not strictly required, as this would require some form of user interface, which is not part of this course.

**Projects realized in VR on a Quest headset are especially appreciated!**

### 2.5 Technical Requirements

| | |
|---|---|
| Platform: | Unity 6, Version 6000.3.xxfx |
| Language: | C# for all scripting and logic. |
| Documentation: | - Strong and self-explanatory names for classes, objects, variables and methods<br>- "In-code" documentation with comments focusing on the "why" of actions<br>- External documentation (MS Word, PDF, MS Visio) for class diagrams, object-hierarchy with components, dependencies, flow-charts and state-machines |

### 2.6 Executability and Functionality

Executability: Only working code is considered for grading.
Concepts alone are nothing without functional implementation.



Functionality: Focus on a few working features and functions, rather than a big plan where nothing works in the end.



## 3 Deliverables

Project code: The reproducible project database with all necessary assets and scripts must be uploaded to the designated area on Ilias (or MS-Teams for large projects).

The submission shall occur with a single compressed file, which contains

- Unity project folders "Assets", "Packages", "ProjectSettings" and "UserSettings"

- Unity project folder "Library", but only with the subfolders
  - "APIUpdater",
  - "PackageManager" and
  - "UIElements",

as well as all other files except for the files
"ArtifactDB", "ArtifactDB-lock", "SourceAssetDB" and "SourceAssetDB-lock".

- All external documentation files (MS-Word, PDF, Visio, etc.)

The compressed files and folders shall be submitted / uploaded as .7Z, .ZIP or .RAR.



The submitted code must NOT contain any lambda expressions or ternary operators!

Presentation: The project shall be presented with an emphasis on all the used data structures and algorithms, and how they are used together. This can be done either within the C# scripts or with a PowerPoint presentation or equivalent.

The presentation can be included in the compressed project submission, or it can be sent / uploaded separately immediately after the presentation.

Project Report: An additional explicit project report is not necessary.

# 4 Presentation

Dates: 2026 June 30th, 08:30 – 17:30 and July 1st, 08:30 – 12:30.
Individual times will be assigned in May 2026.

Format: Single-Person: 10 minutes presentation, and 5 minutes app demo and Q&A.
Two-Person teams: 15 minutes presentation, and 10 minutes app demo and Q&A.

Content:
- Overview of the project concept and key features
- List and explanation of the data structures and algorithms used,
- Object interactions, data exchanges, inter-object method calls

# 5 Grading Criteria

| Category | Description | Weight |
|---|---|---|
| Data Structures | Implementing the correct data structures for the different tasks performed | 15% |
| Algorithms | Selection of algorithms for the correct job, with a working implementation | 20% |
| Data Story | How good and realistic is the chosen problem and the selected solution | 15% |
| Naming, comments | Class-, object-, method- and variable names, comments, formality | 10% |
| Documentation | Out-of-code documentation, PowerPoint slides | 10% |
| Presentation | Style, oral skills, focus, organisation, voice, behaviour | 20% |
| Demonstration | Functionality, performance, does it work at all? | 10% |

The combined grade of DA implementation (60%) and presentation (documentation, presentation and demonstration, 40%) form together 30% to 70% of the total module grade, depending on individual choice.

# 6 Project Guidelines

**Independent Work:** Students may work alone or in pairs of two.

**Academic Integrity:** All C# scripts for grading must be original. The use of external assets is permitted, as long as it was acquired legally and is properly credited.

Public source code, internet and LLMs (like Claude and ChatGPT) may be used for inspiration and partial coding, but not as the only source for the submitted scripts. Code plagiarism is strictly prohibited.

**Support Resources:** Students may use Unity documentation, course materials and solutions, internet panels, question-and-answer websites, LLMs and more.

Additional external resources must be credited.

The help of outside experts is permitted as advisor only – NOT directly contributing any code or assets. Any direct help is strictly prohibited.

**Declaration of compliance** As part of the submission, students must personally sign a declaration of compliance, that they did adhere to the rules as outlined in this document.

Prof. Laszlo Arato